

Variational quantum Hamiltonian engineering

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Estimating the expectation value of a Hamiltonian and performing a Hamiltonian simulation are two fundamental tasks in quantum computation, with their efficiency heavily dependent on the Pauli norm of the Hamiltonian, which sums the absolute values of its Pauli coefficients. In this work, we propose a variational quantum algorithm called variational quantum Hamiltonian engineering (VQHE) to minimize the Pauli norm of a Hamiltonian, thereby reducing the overhead for expectation value estimation and Hamiltonian simulation. We first develop a theoretical framework that transforms the Pauli norm minimization problem into a vector l_1 -norm minimization problem. We then design an appropriate cost function and employ parametrized quantum circuits to variationally minimize this cost function. Numerical experiments demonstrate the effectiveness of VQHE in reducing the Pauli norm for both the Ising Hamiltonian and molecular Hamiltonians. Furthermore, we show that VQHE is compatible with grouping strategies, enabling further reductions in measurement complexity for expectation value estimation. Our results highlight the potential of VQHE to enhance the efficiency of quantum algorithms in near-term quantum devices, offering a promising approach for practical quantum computing applications.

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I. INTRODUCTION

Quantum computing, which leverages the principles of quantum mechanics to perform computations, has emerged as a transformative technology with the potential to solve problems that are intractable for classical computers. Over the past few decades, quantum computing has found applications in diverse fields, including optimization [1,2], chemistry [3,4], security [5,6], and machine learning [7]. At the heart of many quantum algorithms lies the Hamiltonian, an operator that represents the total energy of a quantum system. Two of the most fundamental tasks in quantum computing are the estimation of the expectation value of a Hamiltonian [8,9] and the Hamiltonian simulation [10–13].

The expectation value of a Hamiltonian H with respect to a quantum state $|\psi\rangle$, given by $\langle H \rangle = \langle \psi | H | \psi \rangle$, represents the average energy of the quantum state $|\psi\rangle$ when measured on the basis of the eigenstates of the Hamiltonian H . A common approach to estimating $\langle H \rangle$ involves decomposing the Hamiltonian into a sum of Pauli terms, $H = \sum_i h_i P_i$, where h_i are the coefficients and P_i are Pauli operators, and measuring the expectation value of each term individually $\langle P_i \rangle$. Then we calculate the expectation value by postprocessing $\langle P_i \rangle = \sum_i h_i \langle P_i \rangle$ [14–16]. However, the measurement

cost scales with the number of Pauli terms and the magnitude of their coefficients, making this approach inefficient for large or complex systems. To tackle this problem, one can estimate the value by the probabilistic sampling method [14–16]. Sample the Pauli term P_i of the target Hamiltonian H randomly with a certain probability p_i , and make measurements. The probability is proportional to its absolute value of the coefficient, i.e., $p_i = \frac{|h_i|}{\sum_i |h_i|}$. The desired value can be obtained by postprocessing the measurement results. In this case, the number of total measurement times is determined by the summation of the absolute value of Pauli coefficients, i.e., $N \propto (\sum_{i=1}^L |h_i|)^2$.

Hamiltonian simulation is another fundamental task in quantum computing, which simulates the dynamics of a quantum system. The standard methods for Hamiltonian simulation, such as Trotter-Suzuki decomposition [10,11], are practical for sparse Hamiltonian. The gate count of such a method, which quantifies the complexity of Hamiltonian simulation, depends on the number of Pauli terms in the Hamiltonian. If the system's Hamiltonian is not sparse, then the consumption of the standard method will be too large to be acceptable. To address this problem, Campbell proposed an approach called the quantum stochastic drift protocol (qDrift) [17]. This protocol weights the probability of gates by the corresponding interaction strength in the Hamiltonian, leading to a gate count independent of the number of terms in the Hamiltonian but the summation of the absolute value of Pauli coefficients, i.e., $\sum_{i=1}^L |h_i|$.

The complexity of both tasks—expectation value estimation and Hamiltonian simulation—are dependent on the summation of the absolute value of Pauli coefficients, which is called the *Pauli norm* of Hamiltonian. For a given Hamiltonian

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$H = \sum_{i=1}^{4^n} h_i P_i$, the Pauli norm $\|H\|_p$ is defined by

$$\|H\|_p = \sum_{i=1}^{4^n} |h_i|. \quad (1)$$

It is natural to consider mitigating the complexity of the two tasks by reducing the Pauli norm of the given Hamiltonian. In this work, we proposed a variational quantum algorithm (VQA) [18,19] called variational quantum Hamiltonian engineering (VQHE) to minimize the Pauli norm of the Hamiltonian, and the complexity of the two tasks is reduced correspondingly. VQA is a popular paradigm for near-term quantum applications, which uses a classical optimizer to train parametrized quantum circuits (PQCs) [20] to achieve certain tasks. VQAs have been successfully applied to various tasks, such as searching the ground state [21,22] and the excited state [23,24], quantum classification [25,26], quantum information analysis [27–29], and quantum data compression [30,31]. In previous works [32,33], it was proposed to diagonalize the Hamiltonian by VQA. Such a strategy can reduce the complexity of the expectation value of the Hamiltonian because all the Pauli terms of the diagonalized Hamiltonian commute with each other, and they can be measured simultaneously. However, the VQAs to diagonalize the Hamiltonian are too costly. In the algorithm in Ref. [32], we have to estimate each eigenvalue of the Hamiltonian in every iteration. The nonzero eigenvalue of the Hamiltonian might increase exponentially with the system size, making the algorithm impossible. In Ref. [33], the first step of the algorithm to diagonalize the Hamiltonian is preparing the Hamiltonian thermal state, which is a QMA-hard [34] problem. Thus, the algorithm in Ref. [33] is not practical either.

With the engineered Hamiltonian, the processing of a Hamiltonian such as expectation value estimation and Hamiltonian simulation becomes easier to implement. Specifically, we use a parametrized unitary $U(\theta)$ to reduce the Pauli norm of a given Hamiltonian H , i.e., $\min \|U(\theta)HU(\theta)^\dagger\|_p$. We first develop the theory to convert the Pauli norm optimization problem into the state l_1 -norm minimization problem. Then we design an appropriate cost function and minimize it variationally. The VQHE algorithm outputs the engineered Hamiltonian, whose Pauli norm is minimized. We then display how to apply the engineered Hamiltonian to the tasks of the expectation value estimation and the Hamiltonian simulation. Especially in the task of expectation value estimation, we emphasize that the engineered Hamiltonian is compatible with grouping strategy [21,35–37], which is another method to reduce the measurement times.

We validate the effectiveness of VQHE through numerical experiments on the Ising Hamiltonian and molecular Hamiltonians, showing significant reductions in the Pauli norm. Our results highlight the potential of VQHE to enhance the efficiency of quantum algorithms for expectation value estimation and Hamiltonian simulation, making it a valuable tool for practical quantum computing applications.

II. THEORETICAL FRAMEWORK

In this section, we are going to encode the Hamiltonian into a quantum state and prove that applying a unitary channel on

a Hamiltonian is equivalent to applying a unitary gate on the Hamiltonian state.

For a given n -qubit Hamiltonian H , we can always decompose it into a Pauli basis as

$$H = \sum_{i=1}^{4^n} h_i P_i, \quad (2)$$

where h_i is the real Pauli coefficient, and the $P_i \in \{I, X, Y, Z\}^{\otimes n}$ is an n -qubit Pauli tensor product. The problem of our task is minimizing the Pauli norm of Hamiltonian H by optimizing the unitary U , which can be written as

$$P_{\text{opt}}(H) = \min_U \|UHU^\dagger\|_p. \quad (3)$$

On quantum computers, it is not convenient to apply a unitary channel U on the Hamiltonian H directly. For simplicity, our first step is to encode the Hamiltonian into a quantum state by the following definition.

Definition 1 (Hamiltonian vectorization). For a given n -qubit Hamiltonian $H = \sum_{i=1}^{4^n} h_i P_i$, it can be encoded into a $2n$ -qubit state

$$|H\rangle = \frac{1}{\lambda} (h_1, h_2, \dots, h_{4^n})^T, \quad (4)$$

where $|H\rangle$ is the Hamiltonian state, and the Hamiltonian vector refers to the unnormalized Hamiltonian state, T stands for transpose, and λ refers to the normalization factor, i.e.,

$$\lambda = \sqrt{\sum_{i=1}^{L=4^n} h_i^2}.$$

For example, if the given Hamiltonian is $H = I + 2X + 3Y - 4Z$, then the corresponding Hamiltonian state is $|H\rangle = \frac{1}{\sqrt{30}}(1, 2, 3, -4)^T$. The l_1 -norm of the Hamiltonian state is $\| |H\rangle \|_1 = 10/\sqrt{30}$. The state initialization process can be efficiently executed on a quantum device. It was proven in Ref. [38] that an arbitrary n -qubit, d -sparse quantum state can be deterministically prepared with a circuit depth $\Theta(\log_2(nd))$. Fortunately, the number of Pauli terms of the most meaningful Hamiltonian is polynomial, i.e., $\tilde{O}(n^4)$. Thus the corresponding Hamiltonian state is $\tilde{O}(n^4)$ -sparse, implying it can be prepared efficiently.

The engineered Hamiltonian state can be directly represented as $|H'\rangle = |UHU^\dagger\rangle$. Such a state can be equivalently expressed as an *encoded unitary* V applied on the Hamiltonian state $|H\rangle$. The formal statement is shown in the following theorem.

Theorem 1. For a given Hamiltonian H and unitary U , the vectorized engineered Hamiltonian is $|H'\rangle = |UHU^\dagger\rangle$, which is equivalent to applying a unitary gate V on the Hamiltonian state

$$|UHU^\dagger\rangle = V|H\rangle, \quad (5)$$

with $V = \begin{bmatrix} \langle c_1 | \\ \vdots \\ \langle c_n | \end{bmatrix}$, where $|c_i\rangle = |U^\dagger P_i U\rangle$ is the engineered Hamiltonian state of Pauli P_i .

Proof. For a given Hamiltonian H , we can decompose it into a linear combination of Pauli terms, i.e., $H = \sum_i h_i P_i$. The engineered Hamiltonian by unitary U is $H' = UHU^\dagger = \sum_i h'_i P_i$. The engineered coefficients are obtained by

following

$$h'_i = \frac{1}{2^n} \text{Tr}[UHU^\dagger P_i] \quad (6)$$

$$= \frac{1}{2^n} \text{Tr} \left[\sum_j h_j P_j U^\dagger P_i U \right] \quad (7)$$

$$= \frac{1}{2^n} \text{Tr} \left[\sum_j h_j P_j \sum_m c_{im} P_m \right] \quad (8)$$

$$= \frac{1}{2^n} \sum_{j,m} h_j c_{im} \text{Tr}[P_j P_m]. \quad (9)$$

Since $\text{Tr}[P_j P_m] = 2^n \delta_{jm}$, where P_j and P_m are Pauli matrices, n is the number of qubits. Then,

$$h'_i = \sum_j h_j c_{ij} = \langle c_i | H \rangle, \quad (10)$$

where $|c_i\rangle = |U^\dagger P_i U\rangle$ is the engineered Hamiltonian state of Pauli operator P_i . Here we can construct an operator V ,

$$V = \begin{bmatrix} \langle c_1 | \\ \langle c_2 | \\ \vdots \\ \langle c_{4^n} | \end{bmatrix}, \quad (11)$$

such that

$$|H'\rangle = V|H\rangle. \quad (12)$$

Since unitary U preserves the l_2 -norm, (1) $\langle c_i | c_j \rangle = 1$. Because $\text{Tr}[U P_i U^\dagger U P_j U^\dagger] = \text{Tr}[P_i P_j] = 0$ for $i \neq j$, then $\text{Tr}[\sum_m c_{im} P_m \sum_t c_{jt} P_t] = \sum_{m,t} c_{im} c_{jt} \text{Tr}[P_m P_t] = 0$, since $\text{Tr}[P_m P_t] = 2^n \delta_{mt}$, then it becomes $\sum_m c_{im} c_{jm} = 0$, which implies (2) $\langle c_i | c_j \rangle = 0, \forall i \neq j$. The two properties imply that the matrix V is unitary. ■

This theorem informs us that engineering Hamiltonian is equivalent to applying a unitary V on the Hamiltonian vector. Specifically, from Definition 1, the Hamiltonian state must be normalized, so it is straightforward to have the relation between the Pauli norm of Hamiltonian $\|H\|_p$ and the l_1 -norm of Hamiltonian state

$$\|UHU^\dagger\|_p = \lambda \|V|H\rangle\|_1. \quad (13)$$

Since unitary preserves l_2 -norm, meaning λ is a constant, the problem of minimizing Pauli norm $\|H\|_p$ is equivalent to minimizing l_1 -norm of Hamiltonian state $\| |H\rangle \|_1$, which is

$$P_{\text{opt}}(H) = \lambda \min_V \|V|H\rangle\|_1. \quad (14)$$

Thus, when the optimal unitary V is found to minimize the l_1 -norm of $V|H\rangle$, it is equivalent to the optimal U such that $\|UHU^\dagger\|_p$ is minimized. The specific relation between U and V is shown in Sec. III B.

Note that V is unitary (linear operation), so we can derive the following properties.

Corollary 1. The properties of operation V are as follows:

$$(1) \quad |U_2 U_1 H U_1^\dagger U_2^\dagger\rangle = V_2 V_1 |H\rangle, \quad (15)$$

$$(2) \quad (\alpha V_1 + \beta V_2) |H\rangle = \alpha V_1 |H\rangle + \beta V_2 |H\rangle, \quad (16)$$

$$(3) \quad |U_1 \otimes U_2 H U_1^\dagger \otimes U_2^\dagger\rangle = V_1 \otimes V_2 |H\rangle. \quad (17)$$

Proof. For property (1), we can denote the $\tilde{H} = U_1 H U_1^\dagger$, then $|U_2 U_1 H U_1^\dagger U_2^\dagger\rangle = |U_2 \tilde{H} U_2^\dagger\rangle = V_2 |\tilde{H}\rangle$. Applying the theorem again, $V_2 |\tilde{H}\rangle = V_2 |U_1 H U_1^\dagger\rangle = V_2 V_1 |H\rangle$. For property (2), it is obtained straightforwardly from the linearity of unitary V . For property (3), supposing $H = \sum_{ij} h_{ij} P_i \otimes P_j$, then we have

$$U_1 \otimes U_2 H U_1^\dagger \otimes U_2^\dagger = \sum_{ij} h_{ij} U_1 P_i U_1^\dagger \otimes U_2 P_j U_2^\dagger, \quad (18)$$

where $i, j \in \{1, 2, 3, 4\}$ such that P_i, P_j corresponds to Pauli operators $\{I, X, Y, Z\}$, respectively. From Theorem 1, we have $U P_i U^\dagger = V \tilde{a}_i$, where $\tilde{a}_i$ refers to the basis vector, i.e., only position i in $\tilde{a}_i$ is 1 and other positions are all 0. In other words, $|P_i\rangle = \tilde{a}_i$. Correspondingly, $|H\rangle = \frac{1}{\sqrt{\sum_{ij} h_{ij}^2}} \sum_{ij} h_{ij} \tilde{a}_i \otimes \tilde{a}_j$. Now, we have

$$U_1 \otimes U_2 H U_1^\dagger \otimes U_2^\dagger = \sum_{ij} h_{ij} V_1 \tilde{a}_i \otimes V_2 \tilde{a}_j \quad (19)$$

$$= V_1 \otimes V_2 \sum_{ij} h_{ij} \tilde{a}_i \otimes \tilde{a}_j. \quad (20)$$

If we take Hamiltonian vectorization on both sides, it becomes

$$|U_1 \otimes U_2 H U_1^\dagger \otimes U_2^\dagger\rangle = V_1 \otimes V_2 |H\rangle. \quad (21)$$

Note that unitary is an l_2 -norm-preserving operation, so the normalization factors are canceled out in this equation. ■

III. VARIATIONAL QUANTUM ALGORITHMS

It is well known that due to the nonconvexity of unitary, it is very hard to obtain the optimal unitary to satisfy our requirement as shown in Eq. (14). Here we propose to apply the variational quantum algorithm (VQA) to approach the optimal unitary. The workflow of our algorithm is shown in Fig. 1. At the beginning, we encode the Hamiltonian into a quantum state, randomize the parameters in the PQC, and estimate the cost function by measurement. Then we use a classical computer to optimize the parameters in the PQC and update the parameters correspondingly. Repeat this process until the cost function is minimized.

In this section, we are going to present the details of the cost function and the choice of PQC in the algorithm design. The specific algorithm is also provided as shown in Algorithm 1.

A. Cost function

We have shown that the complexities of many tasks depend on the Pauli norm of the Hamiltonian, such as expectation value estimation [35] and Hamiltonian simulation [17]. We then convert the problem of minimizing the Pauli norm [Eq. (3)] to the problem of minimizing the l_1 -norm of the Hamiltonian vector [Eq. (14)]. It is natural to consider the l_1 -norm of the corresponding Hamiltonian state as the cost function, i.e., $\| |H\rangle \|_1$. However, estimating the l_1 -norm of a quantum state is generally hard on quantum devices. Fortunately, there is a quantum algorithm that can efficiently estimate the sum of the fourth power of elements in the

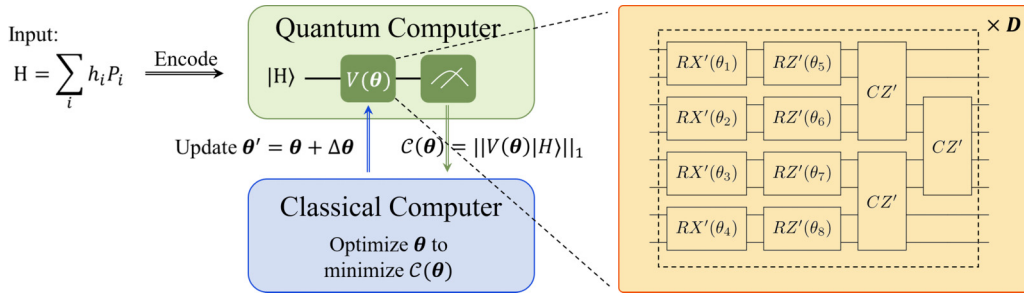


FIG. 1. Diagram of the variational quantum algorithm for Hamiltonian engineering. Encode the given Hamiltonian H into quantum state $|H\rangle$, then apply parametrized quantum circuit $V(\theta)$. Note that the rotation gates $RX'(\theta)$, $RZ'(\theta)$, and control-Z gates CZ' are transferred gates; the exact circuit representation is shown in Fig. 7 in Appendix B. Measuring the circuit and calculating the cost function, which is the l_1 -norm of the quantum state, the classical computer optimizes the cost function by updating the parameters in the PQC. After iterations, the cost function converges to the optimized value.

Hamilton vector $|H\rangle$, i.e.,

$$Q = \sum_{i=0}^{d-1} |\langle i|H\rangle|^4 = \sum_{i=0}^{d-1} |h_i|^4, \quad (22)$$

where d is the dimension of the system. The detail of the algorithm to estimate the quantity Q is shown in the Appendix A. This algorithm is also used in inverse participation ratio estimation [39]. We have that $Q^{\frac{1}{4}} = \| |H\rangle \|_4$, which means when the quantity Q is maximized, then the corresponding l_4 -norm of the Hamiltonian state is maximized, and vice versa. Assuming the unitary that maximized the cost function Q is V^* , then the unitary V^* is also optimal to the problem of maximizing the l_4 -norm of the Hamiltonian state, i.e., $\max_V \|V|H\rangle\|_4 = \|V^*|H\rangle\|_4$. If we apply the optimal unitary V^* to calculate the corresponding 1-norm, we have $\|V^*|H\rangle\|_1$. In the general case, if the l_4 -norm of a normalized vector is larger, then the corresponding l_1 -norm is smaller. Thus, to some extent, maximizing the l_4 -norm is equivalent to minimizing the l_1 -norm. The optimized unitary V^* to maximizing the l_4 -norm can be considered as a feasible solution to the problem of minimizing the l_1 -norm, i.e., $\|V^*|H\rangle\|_1 \geq \min_W \|W|H\rangle\|_1$. Briefly

speaking, when the cost function Q is optimized, then we obtain the upper bound of the original problem (minimizing the l_1 -norm). The bound can be saturated in some cases. Intuitively speaking, both the l_1 -norm and Q describe the uncertainty of a quantum state. From this point of view, minimizing the l_1 -norm and maximizing Q have the same optimizing direction, which reduces the uncertainty of a quantum state. Thus, we set Q as the cost function, i.e.,

$$C = Q = \sum_{i=0}^{d-1} |h_i'|^4. \quad (23)$$

Although minimizing the l_1 -norm and maximizing Q is not exactly the same problem, the effectiveness of setting Q as the cost function emerges in the numerical experiment in Sec. IV D.

B. Parametrized quantum circuit

From Theorem 1, the encoded unitary V is constructed from the unitary U , so U and V have a one-to-one correspondence relation. We can consider unitary V as a function of unitary U , i.e., $V = f(U)$. The function f maps an n -qubit unitary U into a $2n$ -qubit unitary V . Here we take the hardware-efficient ansatz for U as shown in Fig. 2(a), and we automatically obtain the corresponding encoded ansatz V as shown in Fig. 2(b). The ansatz for U contains single qubit rotation gates rotation-X (RX), rotation-Z (RZ), and entangled gates control-Z (CZ). The corresponding encoded unitary components V are denoted as RX' , RZ' , and CZ' , whose detailed information is shown in Appendix B. The encoded unitaries are all real, thus there is no need to worry about the imaginary element in the optimized Hamiltonian state. We need to note that $U(\theta)$ and $V(\theta)$ share the same parameters θ . Once the parameters optimization in unitary $V(\theta)$ is done, we can obtain the unitary $U(\theta)$ directly by inserting the parameters θ into U . The unitary U is necessary when we apply the algorithm to expectation value estimation and Hamiltonian simulation, which will be discussed in Sec. IV.

IV. APPLICATION

In this section, we are going to apply the proposed pre-processing algorithm to reduce the measurement times in the task of expectation value estimation and the gate count in

ALGORITHM 1. Variational quantum Hamiltonian engineering (VQHE).

Input: Classical description of n -qubit Hamiltonian $H = \sum_i h_i P_i$, parametrized quantum circuit (PQC) $V(\theta)$, number of iterations ITR,

Output: The optimized parameters θ^* , engineered Hamiltonian state $|H'\rangle$

- 1: Calculate the normalization factor of the Pauli coefficients
 $\lambda = \sum_i |h_i|$
 - 2: Encode Hamiltonian H into quantum state $|H\rangle$
 - 3: Randomly initialize parameters θ
 - 4: **for** itr = 1, ..., ITR **do**
 - 5: Apply $V(\theta)$ to the Hamiltonian state $|H\rangle$
 - 6: Apply the estimation algorithm (Appendix A) to estimate the cost function in Eq. (23)
 - 7: Maximize the cost function C and update parameter θ
 - 8: **end for**
 - 9: Output the optimized parameters θ^* , and the engineered Hamiltonian state $|H'\rangle$
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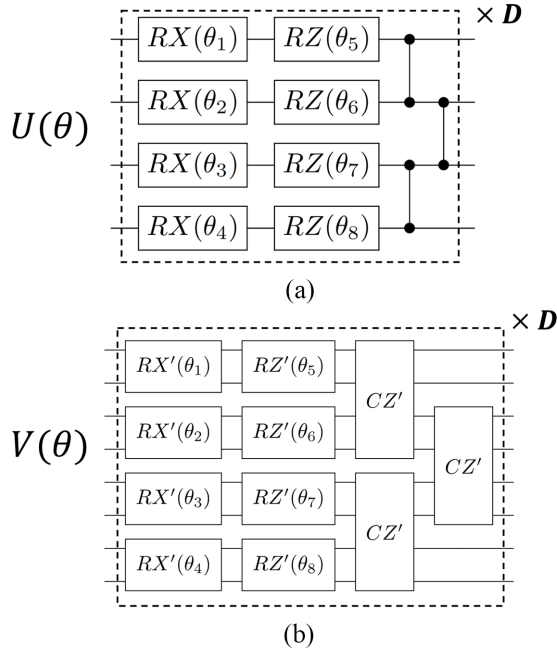


FIG. 2. Structure of entangled layers with D depth. Part (a) is the ansatz $U(\theta)$ for training Hamiltonian H . Part (b) is the ansatz for training Hamiltonian state $|H\rangle$. The circuits for RX' , RY' , and CZ' are displayed in Appendix B.

the task of Hamiltonian simulation, respectively. We especially emphasize that the proposed VQHE is compatible with grouping, which is another method to reduce the measurement complexity. The measurement complexity can be reduced for a further step. We also propose a partition trick to tackle the issue of scalability.

To apply the engineered Hamiltonian H' to the tasks of expectation value estimation and Hamiltonian simulation, we must first obtain the engineered Pauli coefficients h'_i . Quantum state tomography [8] is a candidate to get such information from the engineered Hamiltonian state $|H'\rangle$. However, the cost of state tomography increases exponentially with respect to the system size. Fortunately, the engineered Hamiltonian state $|H'\rangle$ is optimized by reducing its uncertainty, hence the complex amplitude is expected to be concentrated on a certain computational basis. In other words, we expect the engineered Hamiltonian state $|H'\rangle$ to be a sparse pure state. In Ref. [40], a more efficient algorithm than quantum state tomography was proposed to reconstruct the sparse pure state. With the engineered Hamiltonian coefficients h'_i , the l_1 -norm of the engineered Hamiltonian norm can be calculated $\|H'\|_p = \sum_i |h'_i|$ straightforwardly.

A. Expectation value estimation

The expectation value can be estimated by probabilistic sampling strategy. For a given Hamiltonian $H = \sum_i h_i P_i$, where h_i are the real coefficients and P_i are the Pauli operators, we first need to probabilistically sample the Pauli operator from the probability distribution $\{p_i\}$ with $p_i = \frac{|h_i|}{\sum_i |h_i|}$. Then take a measurement on the quantum device, and we denote the outcome as $o^{(1)}$ is either $+1$ or -1 . After sampling N times, we get a list of measurement results $\{o^{(1)}, \dots, o^{(k)}, \dots, o^{(N)}\}$,

and the expectation value can be obtained by

$$\langle H \rangle = \frac{\|H\|_p}{N} \sum_{k=1}^N \text{sgn}(h_i) o^{(k)}, \quad (24)$$

where $\text{sgn}(h_i)$ refers to the sign of h_i , and $N \propto \|H\|_p^2$.

To reduce the measurement times N , we can apply the proposed VQHE algorithm first as shown in Algorithm 1 as a preprocessing, which yields the engineered Hamiltonian state $|H'\rangle$ and the optimized parameters θ^* . Then estimate the engineered Pauli coefficients h'_i , and we arrive at the engineered Hamiltonian $H' = h'_i P_i$.

When we estimate the expectation value of Hamiltonian H , we can estimate the engineered Hamiltonian H' on the engineered state $|\psi\rangle = U|\psi\rangle$ by sampling Pauli terms P_i with probability $p_i = \frac{h'_i}{\sum_i |h'_i|}$, i.e.,

$$\sum_i h'_i \langle \psi | U^\dagger P_i U | \psi \rangle = \langle \psi | U^\dagger H' U | \psi \rangle = \langle \psi | H | \psi \rangle. \quad (25)$$

The unitary U is $U(\theta^*)$, where $U(\theta^*)$ is the ansatz shown in Fig. 2(a), and the parameters θ^* are optimized by Algorithm 1. In this case, the corresponding measurement times of estimating Hamiltonian H reduce to $N \propto \|U(\theta^*) H U^\dagger(\theta^*)\|_p^2$.

The measurement times can be reduced further by applying a grouping algorithm to the engineered Hamiltonian. Grouping [21,35–37] is an efficient classical algorithm to reduce the measurement times in the task of expectation value estimation. The main idea of grouping is dividing the Pauli terms into collections, such that all the Pauli elements in a collection commute with each other, thus they can be measured simultaneously. Many grouping methods have been proposed, such as qubit-wise commuting (QWC) [21], general commuting (GC) [36,37], and sorted insertion [35]. Here we focus on the sorted insertion strategy, which is described as follows (the details of QWC and GC can be found in Appendix C).

For a given Hamiltonian as shown in Eq. (2), the set $\{(h_i, P_i)\}_{i=1}^{4^n}$ is sorted by the absolute value of coefficients h_j so that $|h_1| \geq |h_2| \geq \dots \geq |h_{4^n}|$. Then, in the order $i = 1, \dots, 4^n$, it is checked whether P_i commutes with all elements in an existing collection. If it does, it is added to that collection. If not, a new collection is created and $h_i P_i$ is inserted there. The collections are checked in the order of their creation. After division, there are N collections in total, and the i th collection has m_i Pauli terms. Here we denote the grouped Pauli norm as $\|H\|_{\text{gp}} = \sum_{i=1}^N \sqrt{\sum_{j=1}^{m_i} |h_{ij}|^2}$, which is highly dependent on the group strategy, where h_{ij} is the j th Pauli coefficient in the i th collection. The total number of measurement times of the grouped collections is proportional to the square of the Pauli norm, which is $N \propto \|H\|_{\text{gp}}^2$ [35]. The details and examples are shown in Appendix D. If we apply the VQHE algorithm followed by the sorted grouping algorithm, the corresponding measurement times will be

$$N \propto \|U(\theta^*) H U^\dagger(\theta^*)\|_{\text{gp}}^2. \quad (26)$$

B. Hamiltonian simulation with qDrift

Consider a Hamiltonian $H = \sum_{j=1}^L h_j H_j$, where H_j is Hermitian and normalized, and h_j is the weight. For each H_j , the unitary $e^{-i\tau H_j}$ can be implemented on quantum computers

for any τ . In the Trotter formulas, one divides $U = e^{-iH\tau}$ into segments so that $U = U_r^r$ with $U_r = e^{-iH\tau/r}$ and uses $V_r = \prod_{i=1}^L e^{-i h_j H_j \tau/r}$ to approach U_r in the large- r limit. Repeat this process for r times such that $V_r^r \rightarrow U$ in the large- r limit. It has been proven in Refs. [41–43] that the total gate count of the first order is $O[L^3(\Delta\tau)^2/\epsilon]$, and the $2k$ th order is $O[L^{2+\frac{1}{2k}}(\Delta\tau)^{1+\frac{1}{2k}}/\epsilon^{\frac{1}{2k}}]$, where $\Delta := \max_j h_j$ is the magnitude of the strongest term in the Hamiltonian, and ϵ is the error tolerance. The gate count of this method depends on the terms of decomposition, which is not practical for electronic structure Hamiltonians due to its large number of terms.

The quantum stochastic drift protocol (qDrift) [17] was proposed to solve this problem. Each unitary in the sequence is chosen independently from an identical distribution. Denote $\gamma = \sum_i |h_j|$. The strength τ_j of each unitary is fixed at $\tau_j = t\gamma/G$, which is independent of h_j , so we implement gates of the form $e^{-i\tau H_j}$. Then we randomly choose the unitary $e^{-i\tau H_j}$ with probability $p_j = h_j/\gamma$. The full circuit is described by an ordered list of j values $\mathbf{j} = \{j_1, j_2, \dots, j_N\}$ that corresponds to unitary

$$V_j = \prod_{k=1}^G e^{-i\tau H_{j_k}}, \quad (27)$$

which is selected from the product distribution $P_j = \gamma^{-G} \prod_{k=1}^G h_{j_k}$. The gate count of the Hamiltonian simulation by the qDrift method is $G \sim O[(\gamma t)^2/\epsilon]$, where γ is the Pauli norm of the Hamiltonian.

To reduce the gate count, we can first apply the VQHE algorithm to engineer the Hamiltonian such that the Pauli norm of the engineered Hamiltonian $H' = U(\theta^*) H U^\dagger(\theta^*)$ is minimized. Then, the task of simulating the original Hamiltonian H can be realized by simulating the engineered Hamiltonian H' and sandwiching it by unitaries, which is

$$e^{-iHt} = U^\dagger(\theta^*) e^{-iH't} U(\theta^*). \quad (28)$$

The corresponding gate count of simulating the engineered Hamiltonian is

$$G \sim O[(\gamma' t)^2/\epsilon] + \tilde{O}[U(\theta^*)], \quad (29)$$

where $\gamma' = \|U(\theta^*) H U^\dagger(\theta^*)\|_p$, and $\tilde{O}[U(\theta^*)]$ is the gate count of the unitary, which can be ignored because it is a constant. The proof of Eq. (28) is shown in Appendix E.

C. Scalability

One obstacle to variational quantum algorithms is barren plateaus [44–46], which refers to the phenomenon that the gradient decreases exponentially with the increase of the quantum system. To solve this problem, we propose a straightforward trick called a *partition trick*. Briefly speaking, we can divide the Hamiltonian into parts, where they have common factors. For each part, we extract the common factors and apply the VQHE to the rest of the Hamiltonian. We take the Hamiltonian $H = 3XIXYYZ + 2XIXZZYI - XIXIIZ + IZXIYI + 2IZZIYI$ as an example. We can first divide the Hamiltonian into two parts, and extract the common factor, thus we have $H_1 = XIX \otimes (3YYZ + 2ZYI - IIZ)$ and $H_2 = (IZX + 2IZZ) \otimes IYI$. Then we apply VQHE to minimize $3YYZ + 2ZYI - IIZ$ and $IZX + 2IZZ$, respectively.

The engineered Hamiltonian will be $H' = H'_1 + H'_2$ with $H'_1 = I \otimes U_1 H_1 I \otimes U_1^\dagger$ and similarly $H'_2 = U_2 \otimes I H_2 U_2^\dagger \otimes I$.

The partition trick is compatible with estimating the expectation value. When we estimate the expectation value, we have

$$\langle \psi | H | \psi \rangle = \sum_i \langle \psi | H_i | \psi \rangle = \sum_i \langle \psi | U_i^\dagger H'_i U_i | \psi \rangle, \quad (30)$$

where H_i is the partite sub-Hamiltonian, and $H'_i = U_i H_i U_i^\dagger$ is the engineered sub-Hamiltonian. We should estimate the expectation values of each sub-Hamiltonian first, then sum them up. In such a case, the measurement times are $N \propto \sum_i \|H'_i\|_p^2$. In the task of Hamiltonian simulation, Eq. (28) does not hold anymore, for the unitaries U_i are not the same for different partition Hamiltonian H_i . The partition trick is not compatible with Hamiltonian simulation.

The partition trick gets rid of barren plateaus by dividing a big optimization problem into several small ones, which helps us find the effective unitary to reduce the Pauli norm. However, such a trick will also reduce the expressibility of the ansatz, making us unlikely to obtain the optimal unitary to engineer the Hamiltonian. This trick can be understood as a tradeoff between effectiveness and optimality. In addition, choosing a suitable quantum ansatz and setting the starting parameters carefully can also circumvent barren plateaus to some extent [47].

D. Numerical experiments

In this part, we are going to show the effectiveness of the proposed VQHE, which can reduce the Pauli norm of the Hamiltonian. Specifically, we apply VQHE to the Ising Hamiltonian and the molecules' Hamiltonian. All the numerical simulations in this work are conducted on the Qulacs [48], which is a quantum circuit simulator.

The Ising model, a mathematical model of ferromagnetism in statistical mechanics, involves representing a material as a lattice of discrete points, where each point or “spin” can be in one of two possible states, typically denoted as “up” or “down.” The spins interact with their nearest neighbors, and the interactions are characterized by a coupling constant. The Hamiltonian of the Ising model describes the energy of the system and is given by the sum of the interactions between neighboring spins. The basic form of the Hamiltonian for a one-dimensional Ising model is $H = -J_{ij} \sum_{\{i,j\}} \sigma_i \sigma_j - g_k \sum_k \sigma_k$, where $\{i, j, k\}$ refers to the set.

In the first numerical experiment, we apply the VQHE algorithm to the Ising model with different system sizes, and we compare the original Pauli norm and the engineered Pauli norm. For convenience, we set the coefficients in the Ising Hamiltonian to be the same, i.e., $J_{ij} = g_k = 1$. In this experiment, we consider two Hamiltonians, which are *neighbor Ising Hamiltonian* H_{nei} ($j = i + 1$) and *all-to-all Ising Hamiltonian* H_{all} (i and j are independent), respectively. Specifically, we have

$$H_{\text{nei}} = - \sum_{\{i,i+1\}} Z_i Z_{i+1} + \sum_k X_k, \quad (31)$$

$$H_{\text{all}} = - \sum_{i,j} Z_i Z_j + \sum_k X_k, \quad (32)$$

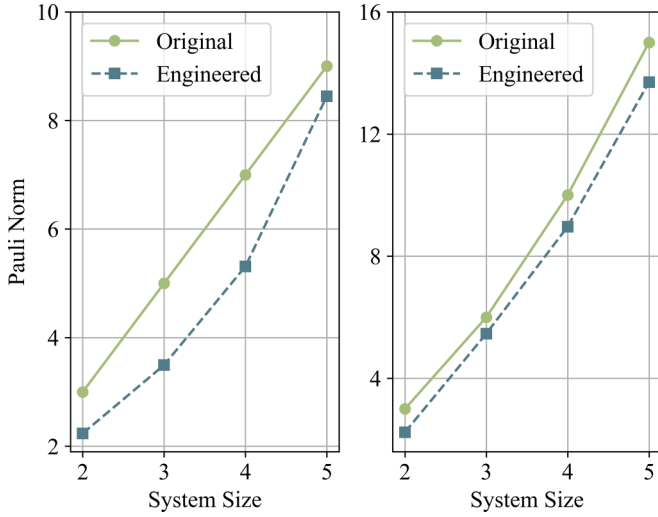


FIG. 3. The Pauli norm comparison of different sizes of neighbor Ising Hamiltonian (left) and all-to-all Ising Hamiltonian (right). The solid line stands for the original Pauli norm, and the dashed line is the engineered Pauli norm.

where X and Z refer to Pauli- X and Pauli- Z operators, respectively. From the numerical results (Fig. 3), the engineered Hamiltonian shows advantages over the original Hamiltonian with smaller Pauli norms in both cases of neighbor Ising Hamiltonian and all-to-all Ising Hamiltonian.

We also conduct numerical experiments on more complicated systems, including H_2 , HeH^+ , H_3^+ , and LiH molecules. In this experiment, we apply VQHE to the tapered Hamiltonian [49,50] of molecules for simplicity. The qubit tapering approach encodes the original quantum system into a smaller system, which preserves the ground state and ground energy. Thus, the tapered Hamiltonian reduces the requirement of the number of qubits, making the variational quantum eigensolver (VQE) [21] more efficient. By applying VQHE to the tapered Hamiltonian of molecules, one observes that the engineered Hamiltonian has a smaller Pauli norm compared with the original one. For a further step, we also take the sorted insertion grouping strategy into consideration. Specifically, we engineer the Hamiltonian with VQHE first, then apply the sorted insertion grouping method to the engineered Hamiltonian. The numerical results in Fig. 4 show that the engineered grouped Pauli norm of Hamiltonian $\|H'\|_{gp}$ can be reduced further compared with that of the engineered Hamiltonian $\|H'\|_p$ and that of the grouped Hamiltonian $\|H\|_{gp}$. These results emphasize that the proposed VQHE is compatible with the grouping strategies.

V. CONCLUSION

In this work, we propose the VQHE algorithm to reduce the Pauli norm of a Hamiltonian such that the overhead for expectation value and Hamiltonian simulation can be reduced. We first develop the theory to convert the Pauli norm optimization problem into the vector l_1 -norm minimization problem, and then design the cost function and parametrized quantum circuits to minimize the Pauli norm variationally. We then display how to apply the proposed VQHE algorithm to

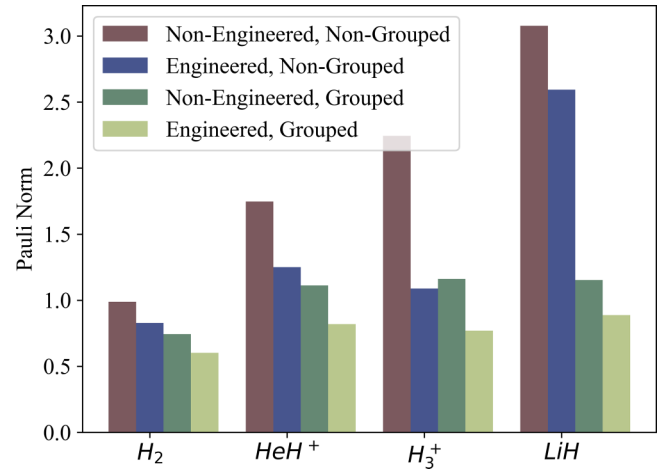


FIG. 4. The Pauli norm of different Hamiltonians of molecules, including H_2 [21], HeH^+ , H_3^+ [51], and LiH [21]. We compare the Pauli norm of the original Pauli norm, the engineered Pauli norm, the grouped Pauli norm [35], and the engineered grouped Pauli norm.

expectation value estimation and Hamiltonian simulation. In the task of expectation value estimation, we also emphasize that the proposed algorithm is compatible with grouping, such that the measurement times can be reduced further. The numerical experiments are conducted by applying the VQHE algorithms to the Ising Hamiltonian and some molecules' Hamiltonian, which shows the effectiveness of the VQHE algorithm.

For further research, it would also be fun to find the optimized Pauli norm of the Hamiltonian we could obtain. It will also be interesting to apply the proposed algorithm VQHE to other applications in quantum computing. In addition, this framework may also be applied to tensor networks, empowering classical computers to solve quantum problems.

ACKNOWLEDGMENTS

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APPENDIX A: QUANTUM ALGORITHM FOR ESTIMATING Q

We denote the summation of the fourth power of elements in a $2n$ -qubit pure state $|\psi\rangle = (h_0, \dots, h_{d-1})^T$ as Q , which is

$$Q = \sum_{i=0}^{d-1} |\langle i|\psi\rangle|^4 = \sum_{i=0}^{d-1} |h_i|^4, \quad (A1)$$

where $|i\rangle$ refers to the computational basis. There exists an efficient quantum algorithm to estimate the quantity Q , which is shown in Fig. 5.

The first step of the algorithm is to initialize the quantum state. We need four registers in total. The first register is

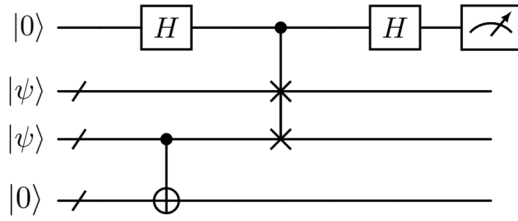


FIG. 5. Quantum circuit for estimating Q . The quantum circuit consists of $2n$ CNOT gates and $2n$ controlled-SWAP gates. Q is estimated through single-qubit measurements on the top qubit.

called the measurement register, which is initialized as $|0\rangle$. The second and third registers are prepared as the $2n$ -qubit quantum state $|\psi\rangle$. The last register is prepared as the $2n$ -qubit zero state $|0\rangle^{\otimes 2n}$. At the beginning, we have

$$\begin{aligned} |\Psi\rangle &= |0\rangle \otimes |\psi\rangle \otimes |\psi\rangle \otimes |0\rangle^{\otimes 2n}, \\ &= \sum_i h_i |0\rangle \otimes |\psi\rangle \otimes |i\rangle \otimes |0\rangle^{\otimes 2n}. \end{aligned} \quad (\text{A2})$$

Applying CNOT gates on the third and fourth register, we then have

$$\text{CNOT}_{3,4}|\Psi_0\rangle = \sum_i h_i |0\rangle \otimes |\psi\rangle \otimes |i\rangle \otimes |i\rangle. \quad (\text{A3})$$

At this step, we can discard the fourth register, and the quantum state on the third register becomes a mixed state $\rho = \sum_i |h_i|^2 |i\rangle\langle i|$. Taking the SWAP test on the first three registers, the probability of getting the measurement as $+1$ is

$$P_{+1} = \frac{1}{2} + \frac{\text{Tr}[\rho|\psi\rangle\langle\psi|]}{2} = \frac{1}{2} + \frac{Q}{2}, \quad (\text{A4})$$

where $Q = \sum_{i=0}^{d-1} |h_i|^4$ is the desired value.

In Ref. [39], a similar algorithm is also used in the inverse participation ratio estimation. Note that the circuit depicted in Fig. 5 requires six times as many qubits as a conventional circuit. Specifically, if we are dealing an n -qubit Hamiltonian, then $6n + 1$ circuits are needed to prepare. However, some techniques such as qubit reset [52] and the destructive SWAP test [53] could be applied to simplify the circuit. If we apply qubit reset [52], then $4n + 1$ qubits are enough as shown in Fig. 6. If the destructive SWAP test [53] is applied to simplify the circuit, then a $4n$ qubit circuit is enough to estimate the cost function Q . Besides, the control SWAP gate can be avoided by the destructive SWAP test. The simplified circuit is more friendly to near-term devices.

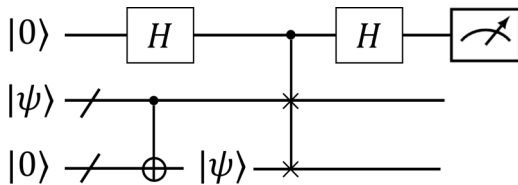


FIG. 6. Quantum circuit for estimating Q with qubit reset simplification. The quantum circuit consists of $2n$ CNOT gates and $2n$ controlled-SWAP gates. Q is estimated through single-qubit measurements on the top qubit.

APPENDIX B: GATE CORRESPONDENCE

The ansatz for U contains single-qubit rotation gates rotation-X (RX), rotation-Z (RZ), and entangled gates control-Z (CZ). The corresponding encoded unitary components V are denoted as RX' , RZ' , and CZ' . The specific unitary matrix representation and the circuit representation are shown in Fig. 7

APPENDIX C: MEASUREMENT TIMES OF GROUPING

Estimating the expectation value of the Hamiltonian is a critical step in many quantum algorithms such as variational quantum eigensolver (VQE) [21]. The expectation value of a Hamiltonian H with respect to a quantum state $|\psi\rangle$ is given by $\langle H \rangle = \langle \psi | H | \psi \rangle$. This quantity represents the average energy of the quantum state $|\psi\rangle$ when measured on the basis of the eigenstates of the Hamiltonian H . For a given Hamiltonian H , we can write it as

$$H = \sum_{i=1}^L h_i P_i, \quad (\text{C1})$$

where L is the number of terms, and h_i is the coefficient of Pauli terms P_i . To estimate the expectation value of Hamiltonian H , one can estimate the expectation value of each Pauli term and sum them up by postprocessing $\langle H \rangle = \sum_i h_i \langle P_i \rangle$, with variance [14,54]

$$\epsilon^2 = \sum_i^L \frac{|h_i|^2 \text{Var}[P_i]}{S_i}, \quad (\text{C2})$$

where S_i are the measurement times of each of the operators P_i , and Var denotes the variance in the measurement of the operator for the given trial state.

Generally, the variance of Pauli $\text{Var}[P_i]$ is unknown in advance. However, since the terms P_i are Pauli, the variance is upper bounded by $\text{Var}[P_i] \leq 1$ [14,54]. One can confirm (e.g., via the use of Lagrange multipliers [16]) that the lowest number of measurements required to bound variance below ϵ^2 can be found by choosing $S_i \propto |h_i|$. This implies that the total number of measurement times N is upper bounded by

$$N \leq \frac{\|H\|_p^2}{\epsilon^2}, \quad (\text{C3})$$

where $\|H\|_p = \sum_i |h_i|$ is called the Pauli norm.

To reduce the total sampling times for expectation value estimation, a grouping strategy is proposed. The main idea is to partition the Pauli terms into groups, and in each group, all elements commute with each other. Since commuting Pauli terms can be measured simultaneously, all elements in a group can be measured at the same time. This kind of method is called *Clique Cover*. If the number of such groups is minimum, it is called *Minimum Clique Cover*.

1. Qubit-wise commuting

Qubit-wise commuting (QWC) [21] is one of the simplest methods to partition the Hamiltonian terms, which only group the terms that commute in qubit-wise. For example, $\{I_1 X_2, Z_1 X_2\}$ is a QWC group because I_1 commutes with X_1 and X_2 commutes with X_2 , while the case of $\{X_1 X_2, Z_1 Z_2\}$ is not a QWC group because X_1 and Z_1 do not commute.

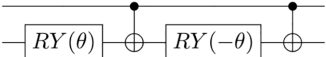
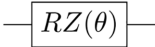
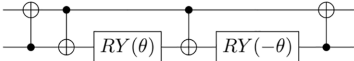
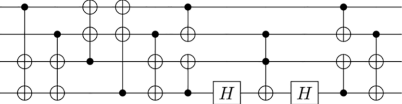
Gate	Unitary	Matrix Representation	Circuit Representation
Rotation-X	$RX(\theta)$	$RX(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$	
	$RX'(\theta)$	$RX'(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \theta & -\sin \theta \\ 0 & 0 & \sin \theta & \cos \theta \end{pmatrix}$	
Rotation-Z	$RZ(\theta)$	$RZ(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$	
	$RZ'(\theta)$	$RZ'(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$	
Control-Z	CZ	$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	
	CZ'	$CZ' = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$	

FIG. 7. Unitary gates U and the corresponding encoded unitary V .

When doing the measurement, we only need to apply for local Clifford unitary U . This method can decrease 70% terms.

2. General commuting

The general commuting (GC) method [36,37] is a more powerful method compared with QWC. Note that QWC is simply the subset of GC corresponding to the case in which the number of noncommuting indices is 0. When we make measurements, we need to apply the unitary U transfer to the GC group into the QWC group such that we can make measurements directly, i.e.,

$$\langle H \rangle = \sum_i \langle \phi | H_i | \phi \rangle \quad (\text{C4})$$

$$= \sum_i \langle \phi | U_i^\dagger U_i H_i U_i^\dagger U_i | \phi \rangle \quad (\text{C5})$$

$$= \sum_i \langle \phi | U_i^\dagger A_i U_i | \phi \rangle \quad (\text{C6})$$

$$= \sum_i \langle \psi | A_i | \psi \rangle = \sum_i \langle A_i \rangle, \quad (\text{C7})$$

where A_i is the corresponding QWC group. The unitary U can be decomposed into Clifford gates. This method can decrease two two-magnitude terms.

After partitioning into GC groups, we need to apply unitaries first, and then make simultaneous measurements. Since the unitaries are Clifford [55], the coefficient will not change after the unitary.

APPENDIX D: THE GROUPED PAULI NORM

For a given operator H , we have

$$H = \sum_{i=1}^N H_i = \sum_{i=1}^N \sum_{j=1}^{m_i} h_{ij} P_{ij}, \quad (\text{D1})$$

where N is the number of collections of mutually commuting operators, m_i is the number of operators in collection i , P_{ij} is the j th Pauli operator in the i th collection, and $h_{ij} \in \mathbb{R}$ is its coefficient.

It has been proved [14,16,56] that the optimal number of m_i measurements of collection i is

$$n_i = \frac{1}{\epsilon^2} \sqrt{\text{Var}[H_i]} \sum_{j=1}^N \sqrt{\text{Var}[H_j]}. \quad (\text{D2})$$

Therefore, the total number of measurements is

$$M = \sum_i^N n_i = \left(\frac{1}{\epsilon} \sum_{i=1}^N \sqrt{\text{Var}[H_i]} \right)^2, \quad (\text{D3})$$

where

$$\text{Var}[H_i] = \text{Cov}[H_i, H_i] = \langle H_i^2 \rangle - \langle H_i \rangle^2. \quad (\text{D4})$$

Theorem D1 [36]. Given M_1, M_2 , two commuting but non-identical Pauli strings, $\mathbb{E}(\text{Cov}[M_1, M_2]) = 0$, where the expectation is taken over a uniform distribution over all possible state vectors.

For convenience, we assume that we do not know the quantum state, and we replace the variances and covariances by their expectation value over a uniform spherical distribution. From Theorem D1, it is straightforward to have

$$M = \sum_i^N n_i = \left(\frac{1}{\epsilon} \sum_{i=1}^N \sqrt{\sum_{j=1}^{m_i} |h_{ij}|^2} \right)^2. \quad (\text{D5})$$

To make a comparison with the ungrouped Pauli norm of the Hamiltonian, we denote it as the grouped Pauli norm

$$\|H\|_{\text{gp}} = \sum_{i=1}^N \sqrt{\sum_{j=1}^{m_i} |h_{ij}|^2}. \quad (\text{D6})$$

Here we give a simple example by setting the Hamiltonian as $H = 3XI - YY + 2ZZ$. The first step is to sort it by the absolute value of coefficients, which will be $\{3XI, 2ZZ, -1YY\}$. At this stage, no collection is created, so we need to create one for the first element, such that $\{3XI\}$. For the second element $2ZZ$, we have to create another collection for ZZ anticommuting to XI . Now the collections will be $\{3XI\}, \{2ZZ\}$. For the third element $-1YY$, it does not commute with the element in

the first collection, thus we need to check the commutativity of the second collection. Luckily, the third element commutes with all elements in the second collection, thus we can put it into this collection, which becomes $\{3XI\}, \{2ZZ, -1YY\}$. In this case, the corresponding grouped Pauli norm is $\|H\|_{\text{gp}} = 5.24$, while the Pauli norm without grouping is $\|H\|_p = 6$.

APPENDIX E: PROOF OF EQ. (28)

When we simulate the engineered Hamiltonian, we have to sandwich it with unitaries to guarantee that the simulated Hamiltonian is the desired one,

$$U^\dagger e^{-iH^\dagger t} U = U^\dagger e^{-iUHU^\dagger t} U \quad (\text{E1})$$

$$= U^\dagger \sum_n \frac{(-it)^n}{n!} (UHU^\dagger)^n U \quad (\text{E2})$$

$$= U^\dagger \sum_n \frac{(-it)^n}{n!} (UHU^\dagger UHU^\dagger \dots UHU^\dagger) U \quad (\text{E3})$$

$$= U^\dagger \sum_n \frac{(-it)^n}{n!} (UH^n U^\dagger) U \quad (\text{E4})$$

$$= U^\dagger U \sum_n \frac{(-it)^n}{n!} (H^n) U^\dagger U \quad (\text{E5})$$

$$= e^{-iHt}. \quad (\text{E6})$$

The second and last equations come from the Taylor expansion.

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- [1] E. Farhi, J. Goldstone, and S. Gutmann, A quantum approximate optimization algorithm, [arXiv:1411.4028](#).
 - [2] M. P. Harrigan, K. J. Sung, M. Neeley, K. J. Satzinger, F. Arute, K. Arya, J. Atalaya, J. C. Bardin, R. Barends, S. Boixo *et al.*, Quantum approximate optimization of non-planar graph problems on a planar superconducting processor, [Nat. Phys.](#) **17**, 332 (2021).
 - [3] S. McArdle, S. Endo, A. Aspuru-Guzik, S. C. Benjamin, and X. Yuan, Quantum computational chemistry, [Rev. Mod. Phys.](#) **92**, 015003 (2020).
 - [4] Google AI Quantum, Collaborators, F. Arute, K. Arya, R. Babbush, D. Bacon, J. C. Bardin, R. Barends, S. Boixo, M. Broughton, B. B. Buckley *et al.*, Hartree-Fock on a superconducting qubit quantum computer, [Science](#) **369**, 1084 (2020).
 - [5] C. H. Bennett and G. Brassard, Quantum cryptography: Public key distribution and coin tossing, [Theor. Comput. Sci.](#) **560**, 7 (2014).
 - [6] A. K. Ekert, Quantum cryptography based on Bell's theorem, [Phys. Rev. Lett.](#) **67**, 661 (1991).
 - [7] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, Quantum machine learning, [Nature \(London\)](#) **549**, 195 (2017).
 - [8] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, Cambridge, 2010).
 - [9] M. Kohda, R. Imai, K. Kanno, K. Mitarai, W. Mizukami, and Y. O. Nakagawa, Quantum expectation-value estimation by computational basis sampling, [Phys. Rev. Res.](#) **4**, 033173 (2022).
 - [10] H. F. Trotter, On the product of semi-groups of operators, [Proc. Am. Math. Soc.](#) **10**, 545 (1959).
 - [11] M. Suzuki, General theory of fractal path integrals with applications to many-body theories and statistical physics, [J. Math. Phys.](#) **32**, 400 (1991).
 - [12] G. H. Low and I. L. Chuang, Optimal Hamiltonian simulation by quantum signal processing, [Phys. Rev. Lett.](#) **118**, 010501 (2017).
 - [13] G. H. Low and I. L. Chuang, Hamiltonian simulation by qubitization, [Quantum](#) **3**, 163 (2019).
 - [14] D. Wecker, M. B. Hastings, and M. Troyer, Progress towards practical quantum variational algorithms, [Phys. Rev. A](#) **92**, 042303 (2015).
 - [15] W. J. Huggins, J. R. McClean, N. C. Rubin, Z. Jiang, N. Wiebe, K. B. Whaley, and R. Babbush, Efficient and noise resilient measurements for quantum chemistry on near-term quantum computers, [npj Quantum Inf.](#) **7**, 23 (2021).
 - [16] N. C. Rubin, R. Babbush, and J. McClean, Application of fermionic marginal constraints to hybrid quantum algorithms, [New J. Phys.](#) **20**, 053020 (2018).
 - [17] E. Campbell, Random compiler for fast Hamiltonian simulation, [Phys. Rev. Lett.](#) **123**, 070503 (2019).
 - [18] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio *et al.*, Variational quantum algorithms, [Nat. Rev. Phys.](#) **3**, 625 (2021).

- [19] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, Quantum circuit learning, *Phys. Rev. A* **98**, 032309 (2018).
- [20] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, Parameterized quantum circuits as machine learning models, *Quantum Sci. Technol.* **4**, 043001 (2019).
- [21] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, and J. M. Gambetta, Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets, *Nature (London)* **549**, 242 (2017).
- [22] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, and J. L. O'Brien, A variational eigenvalue solver on a photonic quantum processor, *Nat. Commun.* **5**, 4213 (2014).
- [23] J. R. McClean, M. E. Kimchi-Schwartz, J. Carter, and W. A. De Jong, Hybrid quantum-classical hierarchy for mitigation of decoherence and determination of excited states, *Phys. Rev. A* **95**, 042308 (2017).
- [24] K. M. Nakanishi, K. Mitarai, and K. Fujii, Subspace-search variational quantum eigensolver for excited states, *Phys. Rev. Res.* **1**, 033062 (2019).
- [25] M. Schuld, A. Bocharov, K. M. Svore, and N. Wiebe, Circuit-centric quantum classifiers, *Phys. Rev. A* **101**, 032308 (2020).
- [26] E. Farhi and H. Neven, Classification with quantum neural networks on near term processors, [arXiv:1802.06002](https://arxiv.org/abs/1802.06002).
- [27] R. Chen, B. Zhao, and X. Wang, Near-term efficient quantum algorithms for entanglement analysis, *Phys. Rev. Appl.* **20**, 024071 (2023).
- [28] R. Chen, Z. Song, X. Zhao, and X. Wang, Variational quantum algorithms for trace distance and fidelity estimation, *Quantum Sci. Technol.* **7**, 015019 (2022).
- [29] X. Zhao, B. Zhao, Z. Wang, Z. Song, and X. Wang, Practical distributed quantum information processing with locnet, *npj Quantum Inf.* **7**, 159 (2021).
- [30] J. Romero, J. P. Olson, and A. Aspuru-Guzik, Quantum autoencoders for efficient compression of quantum data, *Quantum Sci. Technol.* **2**, 045001 (2017).
- [31] C. Cao and X. Wang, Noise-assisted quantum autoencoder, *Phys. Rev. Appl.* **15**, 054012 (2021).
- [32] X. Wang, Z. Song, and Y. Wang, Variational quantum singular value decomposition, *Quantum* **5**, 483 (2021).
- [33] J. Zeng, C. Cao, C. Zhang, P. Xu, and B. Zeng, A variational quantum algorithm for Hamiltonian diagonalization, *Quantum Sci. Technol.* **6**, 045009 (2021).
- [34] Y. Wang, G. Li, and X. Wang, Variational quantum Gibbs state preparation with a truncated Taylor series, *Phys. Rev. Appl.* **16**, 054035 (2021).
- [35] O. Crawford, B. van Straaten, D. Wang, T. Parks, E. Campbell, and S. Brierley, Efficient quantum measurement of Pauli operators in the presence of finite sampling error, *Quantum* **5**, 385 (2021).
- [36] P. Gokhale, O. Angiuli, Y. Ding, K. Gui, T. Tomesh, M. Suchara, M. Martonosi, and F. T. Chong, Minimizing state preparations in variational quantum eigensolver by partitioning into commuting families, [arXiv:1907.13623](https://arxiv.org/abs/1907.13623).
- [37] V. Verteletskyi, T.-C. Yen, and A. F. Izmaylov, Measurement optimization in the variational quantum eigensolver using a minimum clique cover, *J. Chem. Phys.* **152**, 124114 (2020).
- [38] X.-M. Zhang, T. Li, and X. Yuan, Quantum state preparation with optimal circuit depth: Implementations and applications, *Phys. Rev. Lett.* **129**, 230504 (2022).
- [39] Y. Liu, P. Sierant, P. Stornati, M. Lewenstein, and M. Płodzień, Quantum algorithms for inverse participation ratio estimation in multi-qubit and multi-qudit systems, [arXiv:2405.03338](https://arxiv.org/abs/2405.03338).
- [40] B. Gulbahar, K-sparse pure state tomography with phase estimation, [arXiv:2111.04359](https://arxiv.org/abs/2111.04359).
- [41] A. M. Childs, A. Ostrander, and Y. Su, Faster quantum simulation by randomization, *Quantum* **3**, 182 (2019).
- [42] A. M. Childs, D. Maslov, Y. Nam, N. J. Ross, and Y. Su, Toward the first quantum simulation with quantum speedup, *Proc. Natl. Acad. Sci. USA* **115**, 9456 (2018).
- [43] A. M. Childs and Y. Su, Nearly optimal lattice simulation by product formulas, *Phys. Rev. Lett.* **123**, 050503 (2019).
- [44] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, Barren plateaus in quantum neural network training landscapes, *Nat. Commun.* **9**, 4812 (2018).
- [45] C. O. Marrero, Mária Kieferová, and N. Wiebe, Entanglement-induced barren plateaus, *PRX Quantum* **2**, 040316 (2021).
- [46] H.-k. Zhang, C. Zhu, M. Jing, and X. Wang, Statistical analysis of quantum state learning process in quantum neural networks, *Adv. Neural Info. Proc. Sys.* **36**, 33133 (2023).
- [47] X. Liu, G. Liu, H.-K. Zhang, J. Huang, and X. Wang, Mitigating barren plateaus of variational quantum eigensolvers, *IEEE Trans. Quantum Eng.* **5**, 3103719 (2024).
- [48] Y. Suzuki, Y. Kawase, Y. Masumura, Y. Hiraga, M. Nakadai, J. Chen, K. M. Nakanishi, K. Mitarai, R. Imai, S. Tamiya *et al.*, Qulacs: a fast and versatile quantum circuit simulator for research purpose, *Quantum* **5**, 559 (2021).
- [49] S. Bravyi, J. M. Gambetta, A. Mezzacapo, and K. Temme, Tapering off qubits to simulate fermionic hamiltonians, [arXiv:1701.08213](https://arxiv.org/abs/1701.08213).
- [50] K. Setia, R. Chen, J. E. Rice, A. Mezzacapo, M. Pistoia, and J. D. Whitfield, Reducing qubit requirements for quantum simulations using molecular point group symmetries, *J. Chem. Theor. Comput.* **16**, 6091 (2020).
- [51] A. Aguado, O. Roncero, and C. Sanz-Sanz, Three states global fittings with improved long range: singlet and triplet states of H^+_3 , *Phys. Chem. Chem. Phys.* **23**, 7735 (2021).
- [52] D. J. Egger, M. Werninghaus, M. Ganzhorn, G. Salis, A. Fuhrer, P. Müller, and S. Filipp, Pulsed reset protocol for fixed-frequency superconducting qubits, *Phys. Rev. Appl.* **10**, 044030 (2018).
- [53] J. C. Garcia-Escartin and P. Chamorro-Posada, Swap test and Hong-Ou-Mandel effect are equivalent, *Phys. Rev. A* **87**, 052330 (2013).
- [54] J. R. McClean, J. Romero, R. Babbush, and A. Aspuru-Guzik, The theory of variational hybrid quantum-classical algorithms, *New J. Phys.* **18**, 023023 (2016).
- [55] T.-C. Yen, V. Verteletskyi, and A. F. Izmaylov, Measuring all compatible operators in one series of single-qubit measurements using unitary transformations, *J. Chem. Theor. Comput.* **16**, 2400 (2020).
- [56] J. Romero, R. Babbush, J. R. McClean, C. Hempel, P. J. Love, and A. Aspuru-Guzik, Strategies for quantum computing molecular energies using the unitary coupled cluster ansatz, *Quantum Sci. Technol.* **4**, 014008 (2018).